MAC Piping Materials
MAC Piping Materials Sdn. Bhd.
Company No : 1250791-W
Lot 8095, Jalan Sungai Rasau 32,
Kawasan 16, Sungai Rasau, 41300 Klang,
Selangor Darul Ehsan, Malaysia
Tel : +603-3344 9733 Fax : +603-3344 9722
Website : www.macpiping.com
Email : enquiry@macpiping.com

TUV NORD

07/202/9030/WZ/2302/18
07/202/9030/WP/2302/18
PED2014/68/EU

TUV SUD Asia Pacific
ISO 9001 : 2015
Certificate Registration
No. :TUV100 13 4226

# MILL TEST REPORT

## IN ACCORDANCE TO EN 10204 3.1

CERTIFICATE NO : MAC9203283 (1) DATE : 22/11/2023

INVOICE NO. : INVE00873
CONTRACT NO. : SO-1065/1069/1090
SHIPPER : MAC PIPING MATERIALS SDN BHD

PURCHASE ORDER N : 615/005125
CUSTOMER : INTERGROWTH INDUSTRY INC

PRODUCT : STAINLESS STEEL BUTT WELD FITTINGS (SEAMLESS)
SPECIFICATION : ASTM A403/ASME SA403M-19
MATERIAL GRADE : WP304/304L-S (SEAMLESS)

| NO. | DESCRIPTION OF GOODS (PRODUCT, SIZE, SCHEDULE, GRADE) | PO No. | QUANTITY (PCS) | WEIGHT (KGS) | HEAT NUMBER | MECHANICAL TEST Y.S. N/mm² | T.S. N/mm² | EL % (L) | HARDNESS TEST (HRB) | NON-DESTRUCTIVE TEST (NDT) RT | UT | MT | PT | ET | CHEMICAL COMPOSITION (wt%) C | Si | Mn | P | S | Ni | Cr | Mo |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | X 100 | X 1000 | | X 100 | | |
| | | | | | | 205 | 515 | 28 | AVR | | | | | | - | - | - | - | - | 800 | 1800 | - MIN |
| | | | | | | | | | 88 | | | | | | 3.0 | 100 | 200 | 45 | 30 | 1200 | 2000 | - MAX |
| 1 | ELBOW 90 1-1/2" SCH 160S 304/L (S) | 615/005125 | 10 | 7.10 | 33WR305 | 250 | 580 | 48 | 74 | N/A | N/A | N/A | N/A | N/A | 2.1 | 39 | 114 | 31 | 2 | 809 | 1833 | - |
| 2 | ELBOW 90 3" SCH 80S 304/L (S) | 615/005125 | 20 | 60.20 | 33XP321 | 220 | 532 | 42 | 68 | N/A | N/A | N/A | N/A | N/A | 3.0 | 55 | 125 | 30 | 15 | 831 | 1825 | - |
| 3 | ELBOW 90 3/4" SCH 160S 304/L (S) | 615/005125 | 10 | 1.40 | 33TH120 | 295 | 587 | 48 | 72 | N/A | N/A | N/A | N/A | N/A | 2.4 | 51 | 118 | 34 | 2 | 818 | 1819 | - |
| 4 | ELBOW 90 8" SCH 10S 304/L (S) | 615/005125 | 16 | 159.84 | 33DL305 | 276 | 589 | 62 | 77 | N/A | N/A | N/A | N/A | N/A | 2.0 | 36 | 130 | 26 | 2 | 803 | 1818 | - |

ACCEPTED
BY ____ DATE 1/22/21

NOTE :

Y.S = Yield Strength
T.S = Tensile Strength
E.L. = Elongation in 50mm (or 4D), min, %
1Ksi = 1,000 Psi
1Mpa= 1N/mm2 = 0.145038 Ksi

SCH = Schedule
R.T = Radiography Test
U.T = Ultrasonic Test
M.T = Magnetic Practice Test
E.T = Electromagnetic Test
P.T = Dye Penetrant Test

Note
* HEAT TREATMENT : 1080 DEGREE CELSIUS QUENCHED IN WATER (15-30 minute holding time)
* Products conform to ASME B16.9 2018 and ASME Sec II Part A-2019 SA-403/SA-403M 2019.
Cold formed , annealed and quenched, bevelled ends to ASME B16.25,
* Materials are in accordance to NACE MRO175 (ISO 15156-1 & 15156-3 -2015) & NACE MR 0103 -2015.(Latest Edition)
* Intergranular Corrosion Test are in Accordance to ASTM A262 Practice E : Passed
* Surface and dimension check has been carried out 100% to all products
* Pickled and Passivated

* Products are mercury free and free from mercury contamination
* Raw material manufacturer has detected it to be free from radioactive contamination.
* All fittings are confirmed 100% P.M.I tested
* Stainless steelmaking : Electric arc furnace
* Certified according to AD2000 W0 and PED2014/68/EU Annex 1, Paragraph 4.3
* Country of Origin Malaysia

MAC PIPING MATERIALS SDN. BHD.
ACCEPTED
QA/QC DEPARTMENT
QUALITY ASSURANCE

PO: 40324 Unit: 908 HT: 3XP321
PO: 080-25